

Problem #2: Circles and Dots

Solutions:

Assignment Number: Question Instruction: No part marking	Your Answer
2: Among the schemes 6, 7 and 8, which one(s) has/have the same problem as 1, 3 and 5?	6
3: Any prefix code will avoid the problem that schemes 1, 3 and 5 have. TRUE or FALSE?	TRUE
4: For a code to be useful, it must be a prefix code. TRUE or FALSE?	FALSE

Explanation

The codes are not uniquely decodable. That is if a sequence is presented in the code, it is possible to have multiple corresponding words.

Examples (many answers are possible):

Scheme 1: $\bigcirc\bigcirc\odot = D$ or FG

Scheme 3: $\bigcirc\bullet = G$ or BA

Scheme 5: $\odot\bullet\bigcirc\odot\bigcirc\bigcirc = AF$ or DG

Assignment 3:

In a prefix code, if the first n characters code for, let's say X , then there is no other character Y that is coded by the first $n-i$ or $n+i$ characters for all $i > 0$. Hence, the first n character has to represent X . The same logic can be applied, by induction the rest of the string after taking out X . In this way, the whole string can be uniquely decoded.

Assignment 4:

All prefix codes are uniquely decodable, but all uniquely decodable codes are not necessarily prefix codes. For instance scheme 2 is NOT a prefix code (B a prefix of E) but it is uniquely decodable because one can apply the same logic as above but from the end of the string. Since no code is a suffix of another string, such a scheme can also be uniquely decoded.